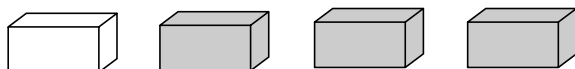


3 points

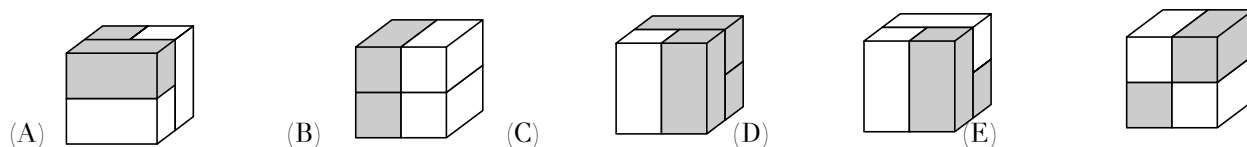
1. Erik has 4 bricks:

Erik 有 4 块砖:



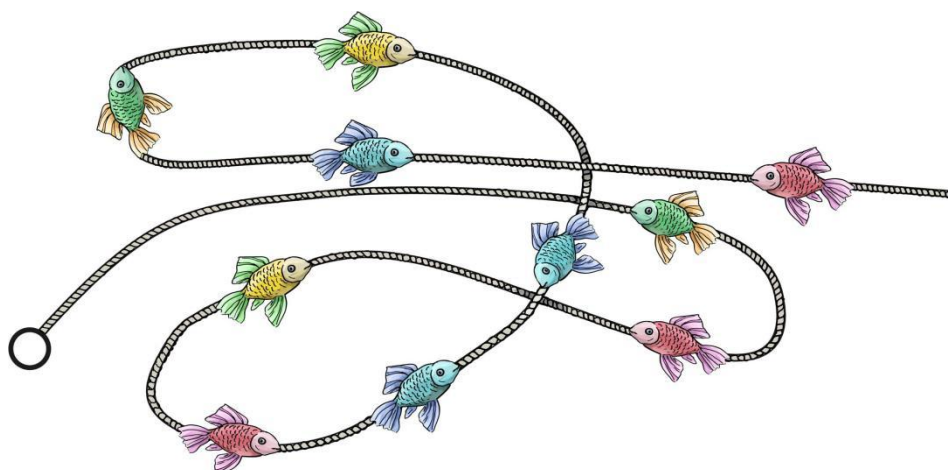
Which of the cubes shown below can he make with his 4 bricks?

用这 4 块砖可以组成以下哪个立方体?



2. How many fish will have their heads pointing towards the ring when we straighten the line?

把下面的鱼线拉直后, 有多少个鱼头是朝着圆环的方向的?



(A) 3 (B) 5 (C) 6 (D) 7 (E) 8

3. When you put the 4 puzzle pieces together correctly, they form a rectangle with a calculation on it. What is the result of this calculation?

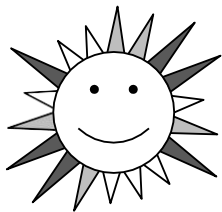
将下列 4 个拼图碎片正确拼接在一起后, 会出现写有一道算式题的矩形。

请问这个算式的答案是多少?



(A) 6 (B) 15 (C) 18 (D) 24 (E) 33

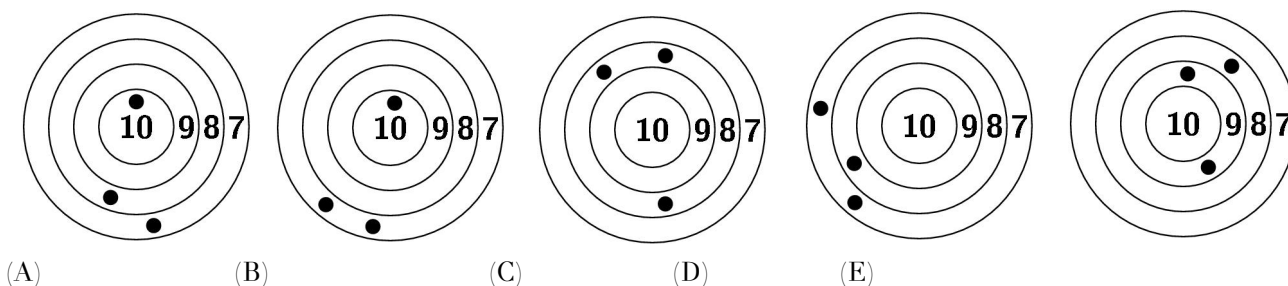
- # 4. Alaya draws a picture of the sun.
Alaya 画了一个太阳。



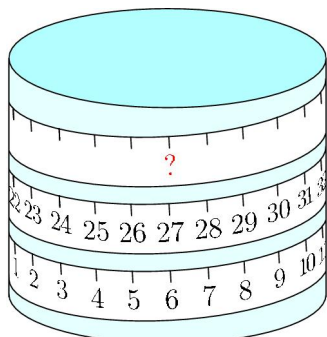
Which of the following answers is part of her picture?
以下哪个选项是这个图画中的一部分？

- (A) (B) (C) (D) (E)

- # 5. Five boys competed in a shooting challenge. Ricky scored the most points. Which target was Ricky's?
5 个男孩进行射击比赛，Ricky 的得分最高。请问下列哪个靶子是 Ricky 的？



- # 6. A measuring tape is wrapped around a cylinder.
Which number should be at the place shown by the question mark?
一把卷尺缠绕在一个圆柱体上。请问图中问号处标记的应该是哪个数？



- (A) 33 (B) 42 (C) 48 (D) 53 (E) 69

7. Denise fired a silver and a gold rocket at the same time.

The rockets exploded into 20 stars in total.

The gold rocket exploded into 6 more stars than the silver one.

How many stars did the gold rocket explode into?

Denise 同时点燃一个银色烟花和一个金色烟花。

两个烟花一共爆炸出 20 颗 “星星”。

其中金色烟花比银色烟花多爆炸出 6 颗 “星星”。

请问金色烟花爆炸出多少颗 “星星”？

(A) 9

(B) 10

(C) 12

(D) 13

(E) 15

8. Rosana has some balls of 3 different colours.

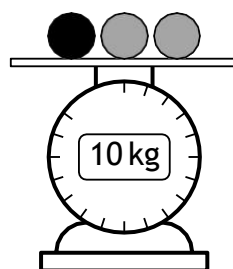
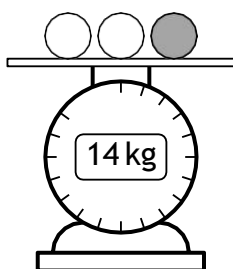
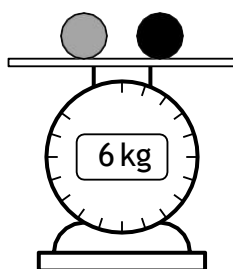
Balls of the same colour have the same weight.

What is the weight of each white ball ?

Rosana 有三种不同颜色的球。

颜色相同的球重量也相同。

请问白球  的重量是多少？



(A) 3 kg

(B) 4 kg

(C) 5 kg

(D) 6 kg

(E) 7 kg

4 points

9. Nisa has 3 different types of cards in a game: apple



cherry



and grapes



She chooses 2 cards from the set and swaps their places.

She wants to arrange the cards so that all the cards with the same fruit on are next to each other.

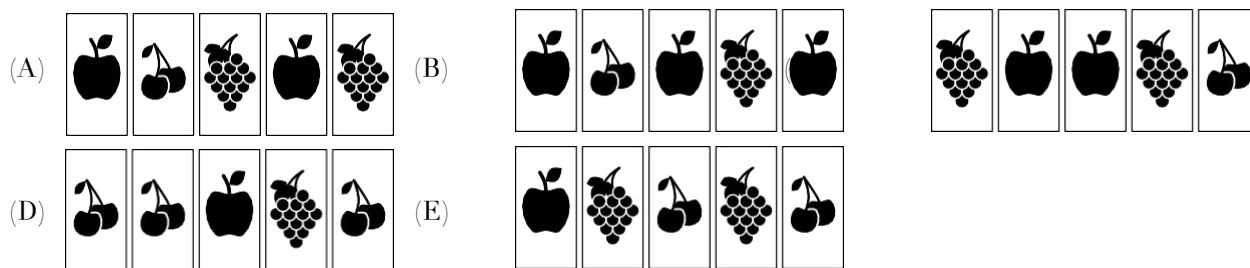
For which set is this **not** possible?



在一场游戏中，Nisa 有三种不同类型的卡片：苹果卡片 ，樱桃卡片  和葡萄卡片 

然后她从以下卡组中选取了两张卡片并且交换了它们的位置，使得有相同水果图案的卡片彼此相邻。

请问以下哪个卡组 **不可能** 实现上述要求？



10. Sofie wants to pick 5 different shapes from the boxes.

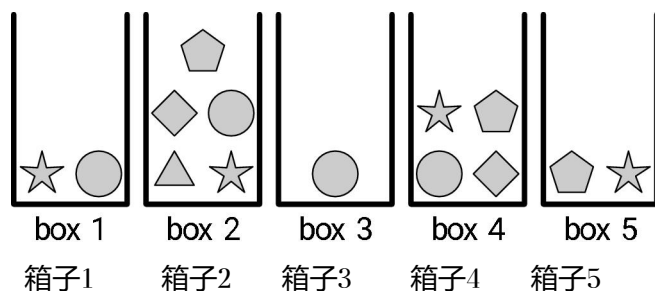
She can only pick 1 shape from each box.

Which shape must she pick from box 4?

Sofie 想要从下面盒子中取出 5 个不同的图形。

她只能从每个箱子中取出一个图形。

请问 Sofie 必须从箱子4 中取出什么图形?



11. 18 cubes are coloured white or grey or black and are arranged as shown.

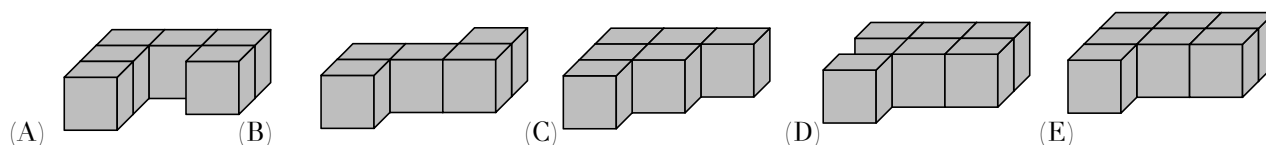
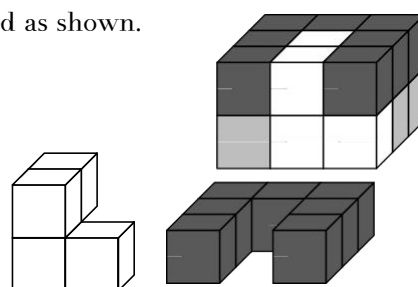
有 18 个白色、灰色或黑色的立方体按右图所示进行组合。

The figures on the right show the white and the black parts.

Which of the following is the grey part?

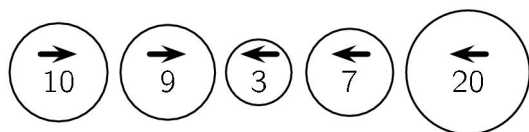
右图显示了其中白色和黑色部分的排列方式。

请问以下哪个选项可以表示灰色部分的排列方式?



12. The 5 balls shown begin to move simultaneously in the directions indicated by their arrows.

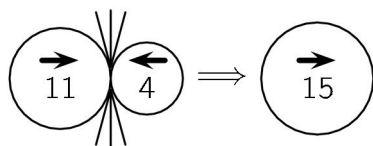
如图所示，5个球按照箭头所指的方向同时移动。



When two balls going in opposite directions collide, the bigger ball swallows the smaller one and increases its value by the value of the smaller ball.

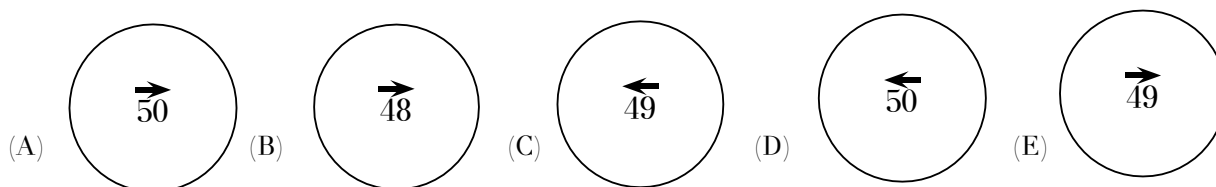
The bigger ball continues to move in its original direction, as shown in the following example.

当两个移动方向相反的球发生碰撞时，大球会吞掉小球，并将小球的值和他本身的值相加。然后大球继续朝着原来的方向移动，如下图所示。



What is the final result of the collisions of the 5 balls shown?

请问上图显示的 5 个球互相碰撞后的最终结果是什么？



13. In an ice cream shop there is some money in a drawer.

After selling 6 ice creams, there are 70 euros in the drawer.

After selling a total of 16 ice creams, there are 120 euros in the drawer.

How many euros were there in the drawer at the start?

在一家冰淇淋店的收银柜里有一些钱。

卖出 6 个冰淇淋之后，收银柜里一共有 70 欧元。

卖出 16 个冰淇淋之后，收银柜里有一共 120 欧元。

请问收银柜里原先有多少欧元？

- (A) 20 (B) 30 (C) 40 (D) 50 (E) 60

14. The Koala ate some leaves from 3 branches. Each branch had 20 leaves.

The Koala ate a few leaves from the first branch and then ate as many leaves from the second branch as were left on the first branch.

Then it ate 2 leaves from the third branch.

How many leaves in total were left on the 3 branches?

考拉吃了三根树枝上的一些叶子。每根树枝上有 20 片叶子。

考拉先后吃了第一根和第二根树枝上的几片叶子。吃的第二根树枝上叶子的数量正好是第一根树枝上吃剩的叶子数量。

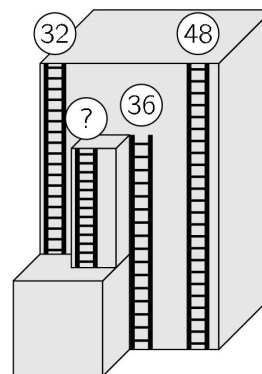
接着考拉又吃了第三根树枝上的 2 片叶子。

请问这三根树枝上一共剩下几片叶子？

- (A) 20 (B) 22 (C) 28 (D) 32 (E) 38

- # 15. On a tall building there are 4 fire escape ladders, as shown.
The heights of 3 ladders are at their tops.
What is the height of the shortest ladder?

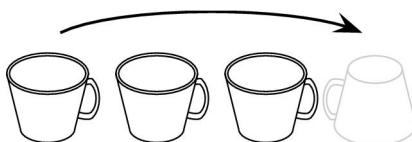
如图所示，在一座高楼中有 4 个消防逃生梯，
其中有 3 个逃生梯的高度标在了梯子顶部。
请问最短的逃生梯高度是多少？



- (A) 12 (B) 14 (C) 16 (D) 20 (E) 22

- # 16. Nora plays with 3 cups on the kitchen table. She takes the left-hand cup, flips it over, and puts it to the right of the other cups.

Nora 在餐桌上玩 3 只杯子。她拿起左手边的杯子，把它翻转过来然后放在了其他杯子的右边。



The picture shows the first move.

What do the cups look like after 10 moves?

上述图片表示的是第一次移动杯子的情况。

请问移动 10 次之后，三只杯子是怎么排列的？

- (A) (B) (C) (D) (E)

5 points

- # 17. Eva has the 5 stickers shown:



She stuck one of them on each of the 5 squares of this board

1	2	3	4	5
---	---	---	---	---

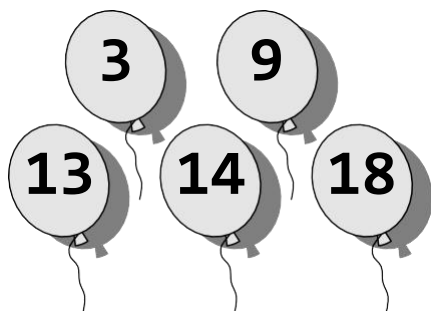
so that is not on square 5, is on square 1, and is adjacent to and .

On which square did Eva stick ?

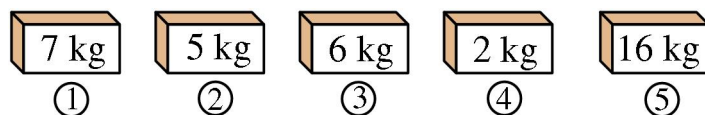
1	2	3	4	5
---	---	---	---	---

请问第几个方格里贴的是 ?

- # 20. Mia throws darts at balloons worth 3, 9, 13, 14 and 18 points.
She scores 30 points in total. Which balloon does Mia *definitely* hit?
Mia 向分值为 3、9、13、14 和 18 分的气球投掷飞镖，
最终她一共得了 30 分。请问她肯定击中了多少分的气球？



- (A) 3 (B) 9 (C) 13 (D) 14 (E) 18
- # 21. A box has fewer than 50 cookies in.
The cookies can be divided evenly between 2, 3, or 4 children.
However, they cannot be divided evenly between 7 children because 6 more cookies would be needed.
How many cookies are there in the box?
一个盒子里的饼干不到 50 块。
这些饼干可以平均分给 2 个、3 个 或 4 个孩子。
如果要平均分给 7 个孩子的话还差 6 块饼干。
请问盒子里一共有多少块饼干？
- (A) 12 (B) 24 (C) 30 (D) 36 (E) 48
- # 22. Each of the 5 boxes contains either apples or bananas, but not both.
The total weight of all the bananas is 3 times the weight of all the apples.
Which boxes contain apples?
在 5 个箱子，有的装着苹果，有的装着香蕉，但不会同时装两种水果。
所有香蕉的总重量是所有苹果总重量的 3 倍。
请问哪个箱子中装的是苹果？



- (A) 1 and 2 箱子 1 和 2 (B) 2 and 3 箱子 2 和 3 (C) 2 and 4 箱子 2 和 4
(D) 3 and 4 箱子 3 和 4 (E) 1 and 4 箱子 1 和 4

23. Elena wants to write the numbers from 1 to 9 in the squares shown.

The arrows always point from a smaller number to a larger one.

She has already written 5 and 7.

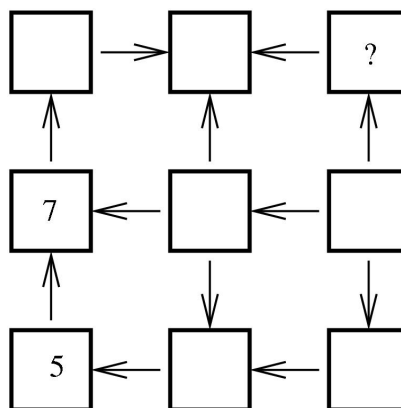
Which number should she write instead of the question mark?

Elena 想把数字 1 到 9 写在下面的方框里。

箭头总是从较小的数指向较大的数。

她已经写了 5 和 7 这两个数。

请问问号处应该填写哪个数？



(A) 2

(B) 3

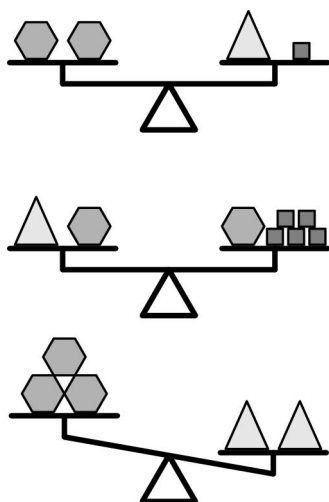
(C) 4

(D) 6

(E) 8

24. Martin placed 3 different types of objects, hexagons , squares  and triangles , on sets of scales, as shown.

如图所示，Martin 在一组天平上面放置了三种不同类型的物品：六边形 ，正方形  和三角形 。



What does he need to put on the left-hand side on the third set of scales for these scales to balance?

他需要在第三个天平的左边再放哪些物品才能让这些天平都保持平衡？

(A) 1 square 1 个正方形

(B) 2 squares 2 个正方形

(C) 1 hexagon 1 个六边形

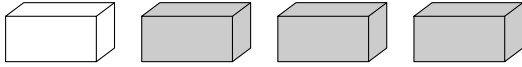
(D) 1 triangle 1 个三角形

(E) 2 triangles 2 个三角形

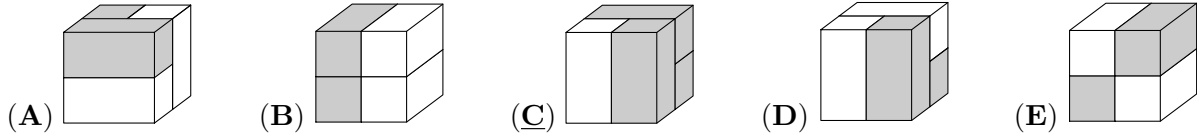
PROBLEMS WITH SOLUTIONS

3 points

1. Erik has 4 bricks:

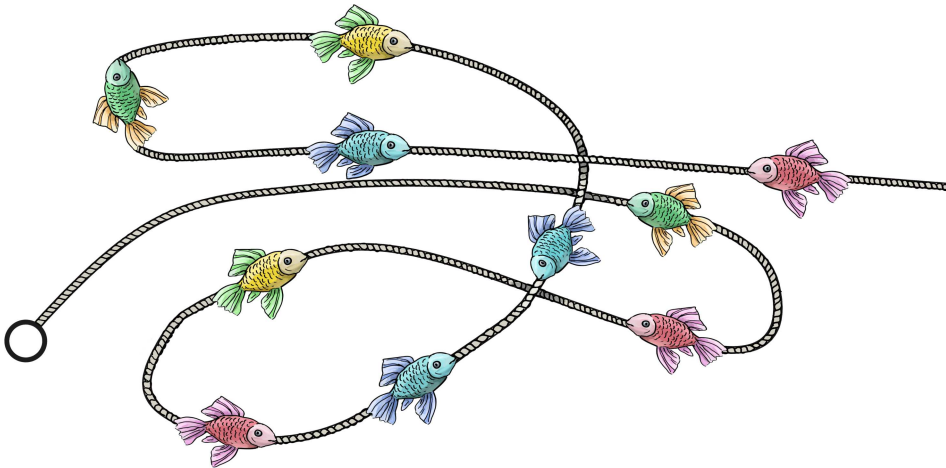


Which of the cubes shown below can he make with his 4 bricks?



SOLUTION:

2. How many fish will have their heads pointing towards the ring when we straighten the line?

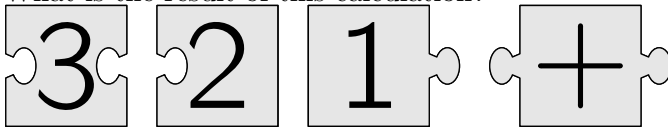


- (A) 3 (B) 5 (C) 6 (D) 7 (E) 8

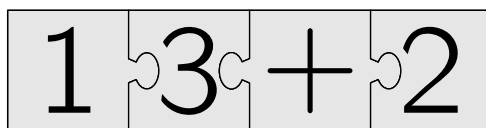
SOLUTION:

3. When you put the 4 puzzle pieces together correctly, they form a rectangle with a calculation on it.

What is the result of this calculation?

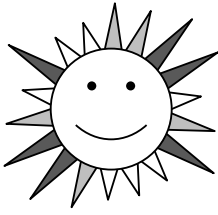


- (A) 6 (B) 15 (C) 18 (D) 24 (E) 33



SOLUTION:

4. Alaya draws a picture of the sun.



Which of the following answers is part of her picture?

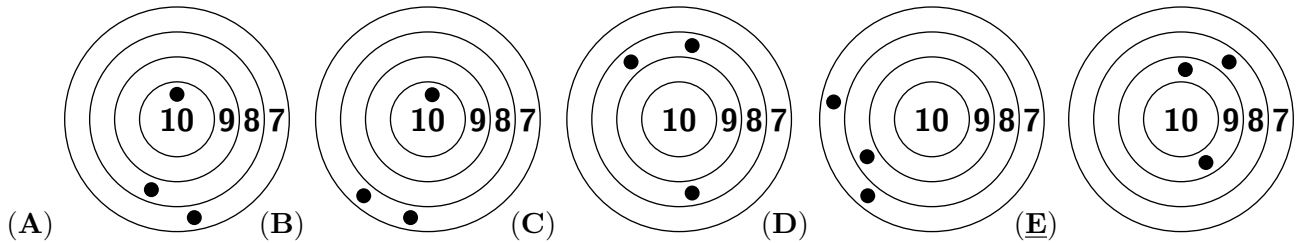
- (A) (B) (C) (D) (E)

SOLUTION:

5. Five boys competed in a shooting challenge.

Ricky scored the most points.

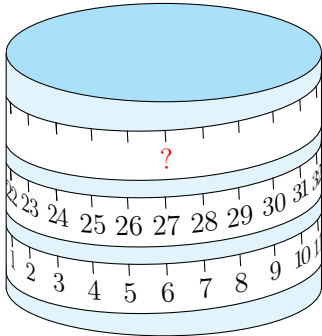
Which target was Ricky's?



SOLUTION: By examining the hits of the 5 targets, we find that target (E) must be Ricky's because the points are 26 in total, vs. 25 in (A), 24 in (B), 24 in (C), 22 in (D).

6. A measuring tape is wrapped around a cylinder.

Which number should be at the place shown by the question mark?



- (A) 33 (B) 42 (C) 48 (D) 53 (E) 69

SOLUTION: The difference between 27 and 6 is 21 and the numbers are increasing moving from bottom to top and from left to right. Hence, the number above 27 must be $27 + 21 = 48$.

7. Denise fired a silver and a gold rocket at the same time.

The rockets exploded into 20 stars in total.

The gold rocket exploded into 6 more stars than the silver one.

How many stars did the gold rocket explode into?

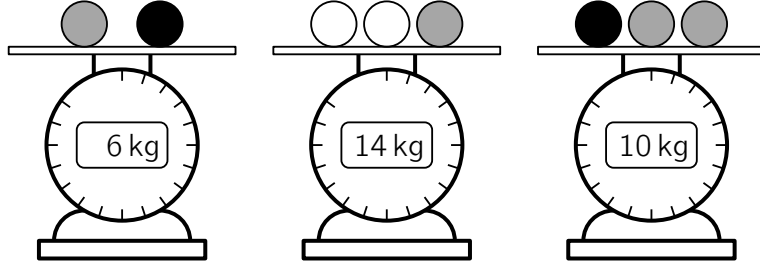
- (A) 9 (B) 10 (C) 12 (D) 13 (E) 15

SOLUTION: The silver rocket had 7 stars, and the gold rocket had 6 more or $7 + 6 = 13$ stars.

8. Rosana has some balls of 3 different colours.

Balls of the same colour have the same weight.

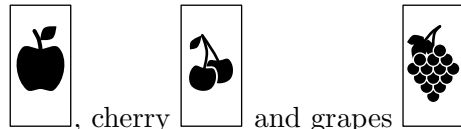
What is the weight of each white ball ?



- (A) 3 kg (B) 4 kg (C) 5 kg (D) 6 kg (E) 7 kg

SOLUTION: $10 \text{ kg} - 6 \text{ kg} = 4 \text{ kg}$ is the weight of one grey/blue ball. Two white/red balls weigh $14 \text{ kg} - 4 \text{ kg} = 10 \text{ kg}$. One white/red ball weighs 5 kg.

4 points



9. Nisa has 3 different types of cards in a game: apple , cherry  and grapes .

She chooses 2 cards from the set and swaps their places.

She wants to arrange the cards so that all the cards with the same fruit on are next to each other.

For which set is this **not** possible?

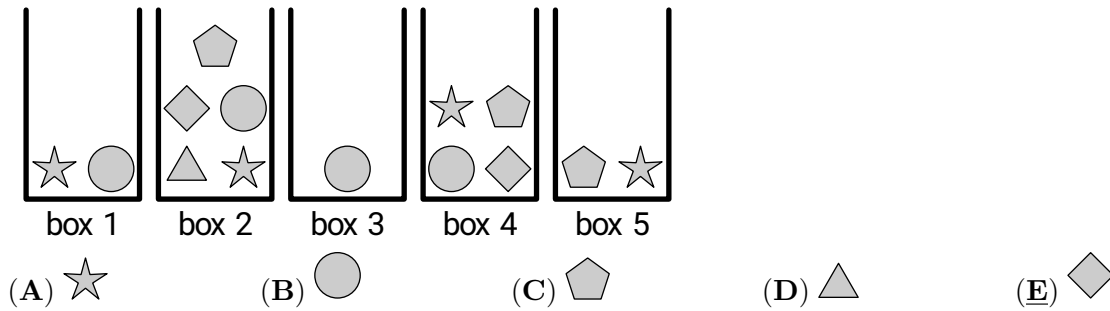
- (A) (B) (C) (D) (E)

SOLUTION: We need to explain by showing how this can be done in 4 of the sets and why not possible in the fifth, with pictures indicating the swaps. In picture (A) we see the two apples are separate and so one of the apples must be moved. For a similar reason we must move one of the grapes. In other words, we must swap one apple and one grape. We have several choices which we test in turn. A) No point swapping the left apple with the left grape as the later does not go next to the other grape. B) No point swapping the left apple with the right grape for the same reason as before. C) No point swapping the right apple with the either grape as this will not bring the apples next to each other. To conclude, this situation will not succeed. All the others have a way to succeed, as seen in the pictures.

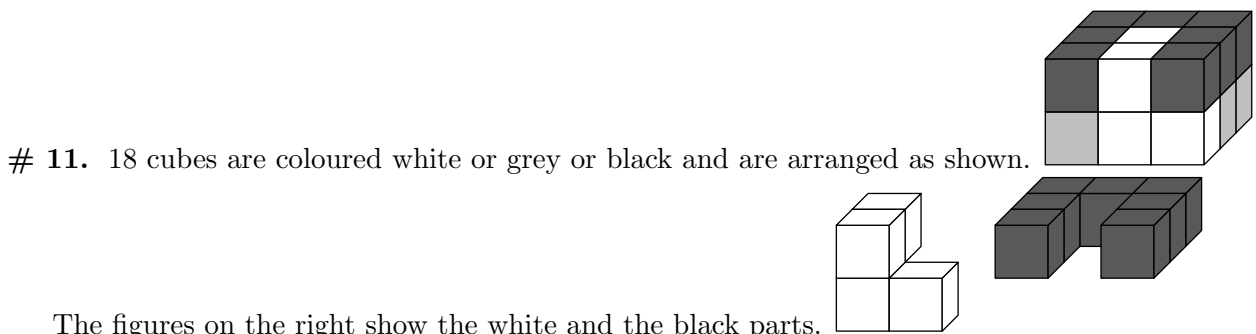
10. Sofie wants to pick 5 different shapes from the boxes.

She can only pick 1 shape from each box.

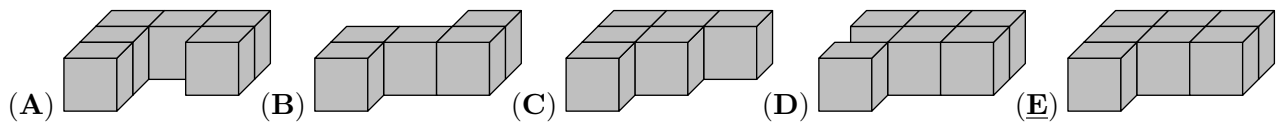
Which shape must she pick from box 4?



SOLUTION: The picking happens in the following order: ● from box 3; ★ from box 1; ⬠ from box 5; ◆ from box 4; ▲ from box 2.

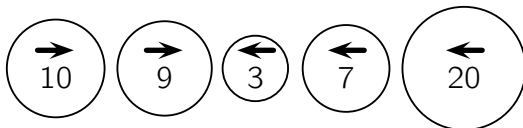


Which of the following is the grey part?



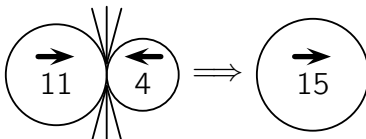
SOLUTION: The grey part consists of 7 cubes in the bottom layer - only two cubes from the entire layer are missing because of the 2 white cubes at the front right.

12. The 5 balls shown begin to move simultaneously in the directions indicated by their arrows.

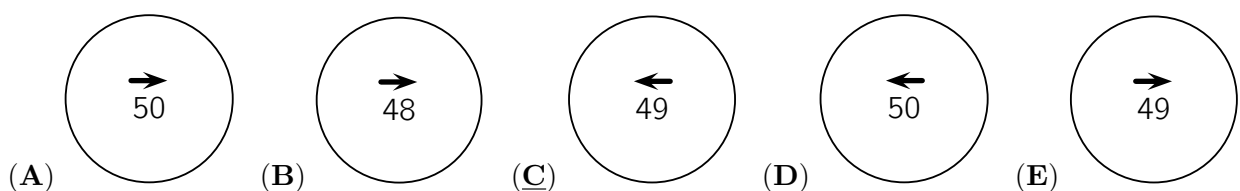


When two balls going in opposite directions collide, the bigger ball swallows the smaller one and increases its value by the value of the smaller ball.

The bigger ball continues to move in its original direction, as shown in the following example.



What is the final result of the collisions of the 5 balls shown?



SOLUTION: 1: 10 - 12 = -2; 2: 10 - 19 = -9; 3: 10 - 39 = -29; 4: -49. The answer is 49 pointing to the left.

- # 13. In an ice cream shop there is some money in a drawer.
 After selling 6 ice creams, there are 70 euros in the drawer.
 After selling a total of 16 ice creams, there are 120 euros in the drawer.
 How many euros were there in the drawer at the start?

(A) 20 (B) 30 (C) 40 (D) 50 (E) 60

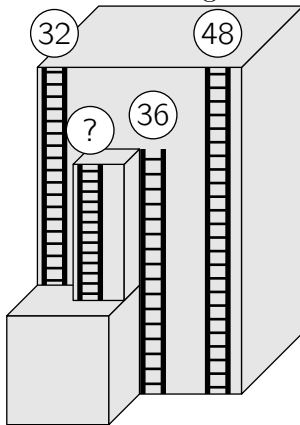
SOLUTION: 10 (= 16 - 6) ice creams cost $120 - 70 = 50$ euros, meaning that 1 ice cream costs 5 euros. Hence, there were $6 \times 5 = 30$ euros less before selling the 6 ice creams, or 40 euros.

- # 14. The Koala ate some leaves from 3 branches.
 Each branch had 20 leaves.
 The Koala ate a few leaves from the first branch and then ate as many leaves from the second branch as were left on the first branch.
 Then it ate 2 leaves from the third branch.
 How many leaves in total were left on the 3 branches?

(A) 20 (B) 22 (C) 28 (D) 32 (E) 38

SOLUTION: The Koala ate 20 leaves in total from the first and second branches, leaving 20 leaves on them. The Koala also ate 2 leaves from the third branch, leaving 18 leaves on it. Or, the Koala left $20 + 18 = 38$ leaves in total.

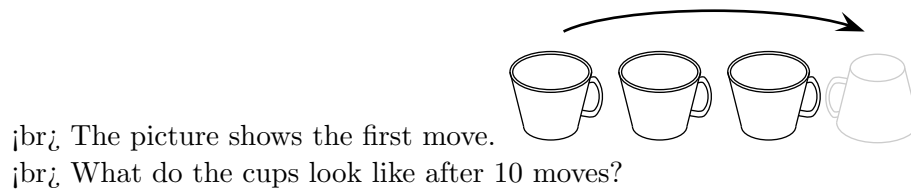
- # 15. On a tall building there are 4 fire escape ladders, as shown.
 The heights of 3 ladders are at their tops.
 What is the height of the shortest ladder?



(A) 12 (B) 14 (C) 16 (D) 20 (E) 22

SOLUTION: Looking at the two ladders on the right we realize that from the top of the shorter ladder to the top of the building it is $48 - 36 = 12$ m. Now look at the two ladders on the left. Subtracting the 12 m just found from the left ladder, we get the height of the short one, namely $32 - 12 = 20$ m.

- # 16. Nora plays with 3 cups on the kitchen table. She takes the left-hand cup, flips it over, and puts it to the right of the other cups.



- (A) (B) (C) (D) (E)

SOLUTION: The first 3 moves put the cups down, next 3 - up, then next 3 - down again. The 10th move will result in the first cups down and the third cup up.

5 points

17. Eva has the 5 stickers shown:

She stuck one of them on each of the 5 squares of this board

1	2	3	4	5
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so that is not on square 5, is on square 1, and is adjacent to and .

On which square did Eva stick ?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

SOLUTION: is in square 1; must be in square 2 because the remaining 3 stickers occupy 3 neighboring squares; hence, must be in square 4 surrounded by and .

18. 7 cards are arranged as shown.

Each card has 2 numbers on with 1 of them written upside down.

The teacher wants to rearrange the cards so that the sum of the numbers in the top row is the same as the sum of the numbers in the bottom row.

She can do this by turning one of the cards upside down.

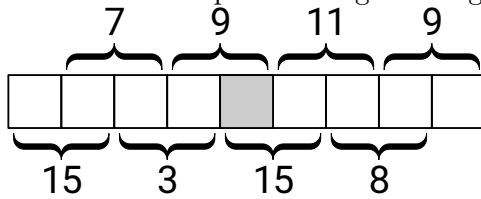
Which card must she turn?

7	5	4	2	8	3	2
7	8	9	9	7	7	7
A	B	C	D	E	F	G

- (A) A (B) C (C) D (D) F (E) G

SOLUTION: Sum at the top is 31 and at the bottom 35, so we want each to become 33. In other words we want to increase the top line by 2 and decrease the bottom by 2. We achieve this by overturning card G (and none other).

- # 19. The numbers 1 to 9 are placed in the squares shown with a number in each square. The sums of all pairs of neighbouring numbers are shown.



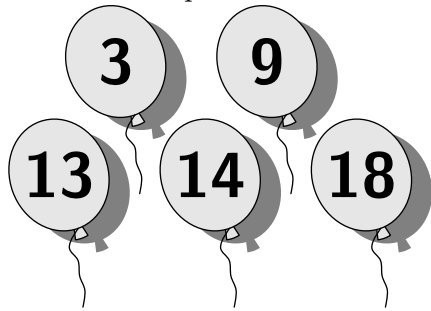
Which number is placed in the shaded square?

- (A) 4 (B) 5 (C) 6 (D) 7 (E) 8

SOLUTION: Method 1: The sum 3 given (bottom row) implies that the numbers in the two relevant squares are 1 and 2, in some order. We try both. It is easy to fill the rest if the numbers (uniquely). The will give 9-6-1-2-7-8-3-5-4 (acceptable) or 10-5-2-1-8-7-4-4-5 (unacceptable). The middle number is 7.

Method 2: The sums of all numbers is $1 + 2 + 3 + \dots + 9 = 45$. By the information given, the sums at the bottom add up to $15 + 3 + 15 + 8 = 41$, but we have one square which did not take into account (the last one). So the last square contains the number $45 - 41 = 4$. We can now easily fill the rest if the numbers (uniquely). The middle number will be seen to be 7.

- # 20. Mia throws darts at balloons worth 3, 9, 13, 14 and 18 points. She scores 30 points in total. Which balloon does Mia *definitely* hit?



- (A) 3 (B) 9 (C) 13 (D) 14 (E) 18

SOLUTION: The score of 30 points can be obtained in two different ways: $3 + 9 + 18$ and $3 + 13 + 14$. Mia definitely hit 3.

- # 21. A box has fewer than 50 cookies in.

The cookies can be divided evenly between 2, 3, or 4 children.

However, they cannot be divided evenly between 7 children because 6 more cookies would be needed.

How many cookies are there in the box?

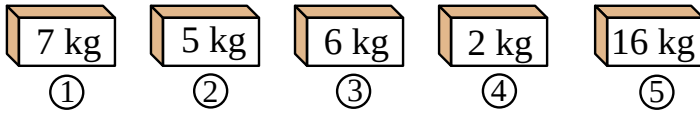
- (A) 12 (B) 24 (C) 30 (D) 36 (E) 48

SOLUTION: The number of cookies that can be divided evenly between 2, 3, or 4 children could only be equal to 48, 36, 24, and 12. Adding 6 more cookies gives 54, 42, 32, and 18. Only 42 cookies can be divided evenly between 7 children. The answer is 36.

- # 22. Each of the 5 boxes contains either apples or bananas, but not both.

The total weight of all the bananas is 3 times the weight of all the apples.

Which boxes contain apples?



- (A) 1 and 2 (B) 2 and 3 (C) 2 and 4 (D) 3 and 4 (E) 1 and 4

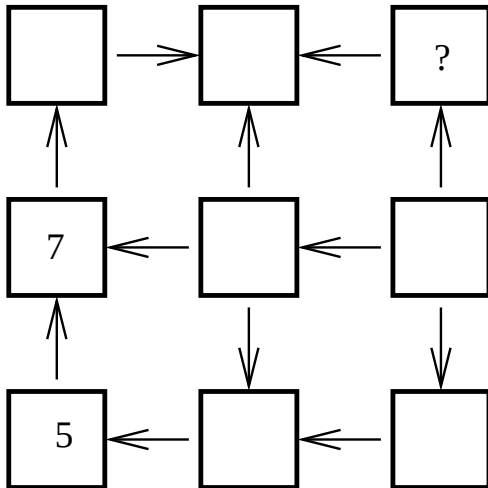
SOLUTION: The total weight is 36 kg; hence, the bananas must weigh 27 kg, and the apples 9 kg. Only box 1 and 4 have combined weight of 9 kg.

23. Elena wants to write the numbers from 1 to 9 in the squares shown.

The arrows always point from a smaller number to a larger one.

She has already written 5 and 7.

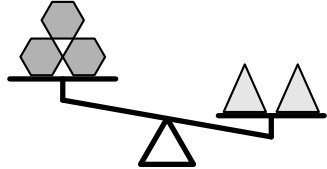
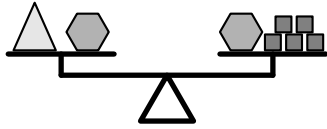
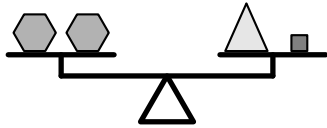
Which number should she write instead of the question mark?



- (A) 2 (B) 3 (C) 4 (D) 6 (E) 8

SOLUTION: 1) 8 must be above 7 and 9 next to it; 2) the number below ? must be 1 because all arrows are pointing out of it and out of its adjacent empty squares. 3) since 5 is in the bottom left square, 6 cannot be placed in any of the empty squares of the second and third rows. 4) 6 must be in the square with the question mark. The numbers could be: Option 1: first row: 8 9 6; second row: 7 2 1; third row: 5 4 3. Option 2: first row: 8 9 6; second row: 7 3 1; third row: 5 4 2.

24. Martin placed 3 different types of objects, hexagons , squares  and triangles , on sets of scales, as shown.



What does he need to put on the left-hand side on the third set of scales for these scales to balance?

- (A) 1 square (B) 2 squares (C) 1 hexagon (D) 1 triangle (E) 2 triangles

SOLUTION: From the balanced middle scale we conclude that the weight of 1 triangle is equal to the weight of 5 squares. Then, from the balanced top scale it follows that the weight of 2 hexagons is equal to the weight of 6 squares, i.e. the weight of 1 hexagon is equal to the weight of 3 squares. Now the bottom scale has 2 triangles on the right, weighing as 10 squares, and 3 hexagons weighing as 9 squares. So we need to put 1 square on the left-hand side in order to balance the scale.